

Chapter: Introduction

Ahonakpon Guy Gbaguidi

Exercise 1.14

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Solution

For any real square matrix $\mathbf{W}$, define

$$\mathbf{W}^S = \frac{\mathbf{W} + \mathbf{W}^T}{2}, \quad \mathbf{W}^A = \frac{\mathbf{W} - \mathbf{W}^T}{2}.$$

In terms of their elements,

$$w_{ij}^S = \frac{w_{ij} + w_{ji}}{2}, \quad w_{ij}^A = \frac{w_{ij} - w_{ji}}{2}.$$

These definitions give

$$\begin{aligned} w_{ji}^S &= \frac{w_{ji} + w_{ij}}{2} = w_{ij}^S, \\ w_{ji}^A &= \frac{w_{ji} - w_{ij}}{2} = -w_{ij}^A, \\ w_{ij}^S + w_{ij}^A &= \frac{w_{ij} + w_{ji}}{2} + \frac{w_{ij} - w_{ji}}{2} = w_{ij}. \end{aligned}$$

Thus, $\mathbf{W}^S$ is symmetric, $\mathbf{W}^A$ is antisymmetric, and $\mathbf{W} = \mathbf{W}^S + \mathbf{W}^A$. In particular, $w_{ii}^A = 0$ for every diagonal element.

Consider the contribution of the antisymmetric part to the quadratic form:

$$\begin{aligned} \sum_{i=1}^D \sum_{j=1}^D w_{ij}^A x_i x_j &= \sum_{i=1}^D \sum_{j=1}^D w_{ji}^A x_j x_i \quad (\text{rename the dummy indices } i \text{ and } j) \\ &= - \sum_{i=1}^D \sum_{j=1}^D w_{ij}^A x_i x_j \quad (\text{antisymmetry and } x_i x_j = x_j x_i). \end{aligned}$$

Let the original double sum be A . The equation above gives $A = -A$, so $2A = 0$ and therefore

$$\sum_{i=1}^D \sum_{j=1}^D w_{ij}^A x_i x_j = 0.$$

Thus, only the symmetric part contributes:

$$\begin{aligned} \sum_{i=1}^D \sum_{j=1}^D w_{ij} x_i x_j &= \sum_{i=1}^D \sum_{j=1}^D (w_{ij}^S + w_{ij}^A) x_i x_j \quad (\text{matrix decomposition}) \\ &= \sum_{i=1}^D \sum_{j=1}^D w_{ij}^S x_i x_j \quad (\text{the antisymmetric contribution is zero}). \end{aligned}$$

It remains to count the independent parameters in the symmetric matrix $\mathbf{W}^S$. There are D independent diagonal elements. For the off-diagonal elements, symmetry gives $w_{ij}^S = w_{ji}^S$, so only one element from each pair is independent. There are $D(D-1)/2$ such pairs. Hence, the total number of independent parameters is

$$D + \frac{D(D-1)}{2} = \frac{D(D+1)}{2}.$$

Therefore, the quadratic form can be represented using only $D(D+1)/2$ independent coefficients. Explicitly,

$$\sum_{i=1}^D \sum_{j=1}^D w_{ij} x_i x_j = \sum_{i=1}^D w_{ii}^S x_i^2 + 2 \sum_{1 \leq i < j \leq D} w_{ij}^S x_i x_j.$$

The factor of 2 accounts for the two equal off-diagonal contributions, so the coefficients can be chosen symmetric without loss of generality.